User Guide

Optical System Design

Instructions, typeset formulas and UI screenshots. Numerical validation remains limited to each analysis' documented scope.

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How calculations work

From prescription to results

Understand the input snapshot, calculation stages and settings behind a result.

1 Read one project snapshot

An analysis captures the selected configuration, aperture, fields, wavelengths, materials and settings. Editors, layouts, formal analyses and laboratories use the shared C# computation core. Display meshes, colors and window size do not replace the prescription. Resynchronize results after editing the project.

2 Generate rays and evaluate

Pupil samples are generated for the selected fields and wavelengths, aimed, intersected, aperture-tested, refracted/reflected and accumulated in optical path. Spot, wavefront, diffraction, energy and optimization calculations then use their own reference and sampling conventions. Two values called RMS are not automatically comparable.



Spot diagram: fields, wavelengths and results

How calculations work

Units, coordinates and dispersion

Track length, angles, wavelengths, local frames and glass models.

Input units and frames

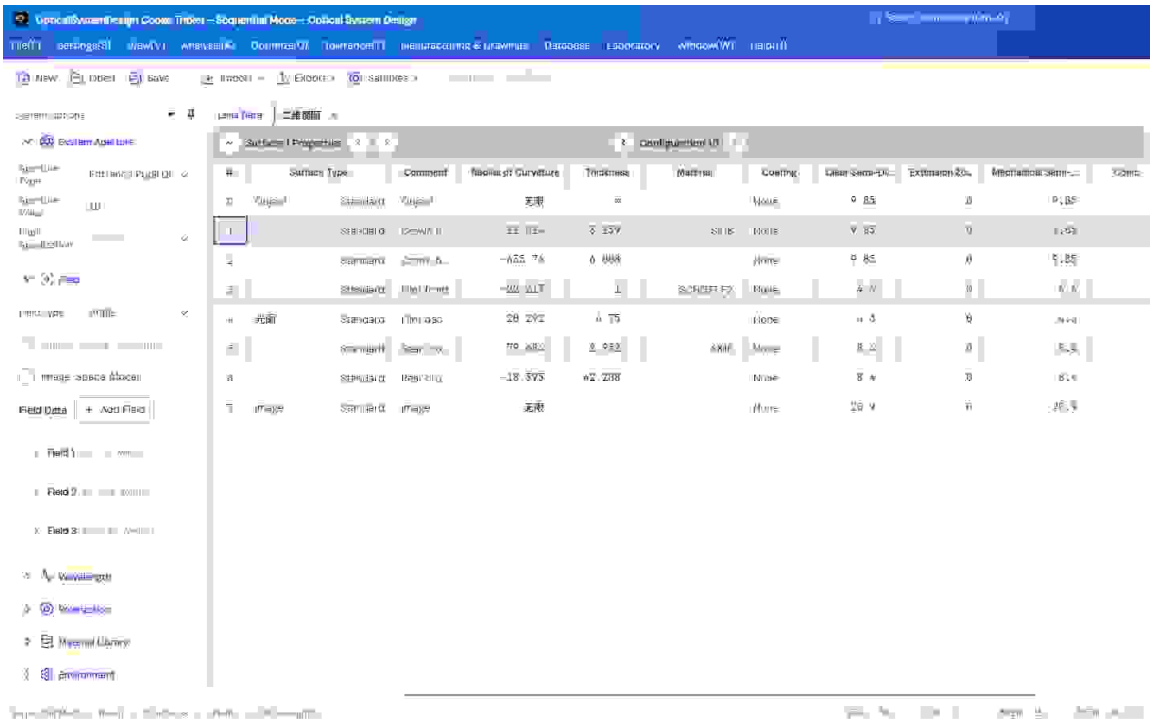
Geometry normally uses mm and material queries use nm; some analysis interfaces/displays use µm, with the conversion shown below. Angular fields and object/image heights are different inputs. Intersections use local surface frames and transform back to global coordinates, preserving decenter, tilt and folded paths. Positive infinite Object thickness means an infinite conjugate; zero remains a finite coincident object.

Index and extinction

Materials evaluate n(λ) and k(λ) from the stored model: catalog dispersion, Sellmeier, Cauchy or tabulated interpolation. The example is a Sellmeier form, usually with λ in µm; coefficient units follow the model. Do not substitute coefficients from another dispersion formula. A shared name does not prove identical physics: check catalog, wavelength range and provenance.

$$n^2(\lambda) = 1 + \sum_j \frac{B_j \lambda^2}{\lambda^2 - C_j}$$

$$1\,\mathrm{mm} = 10^3\,\mu\mathrm{m} = 10^6\,\mathrm{nm}$$



Actual main-workbench screenshot

How calculations work

Real-ray intersections, refraction and reflection

Trace the actual surfaces and accumulate geometric and optical path lengths.

Intersections and apertures

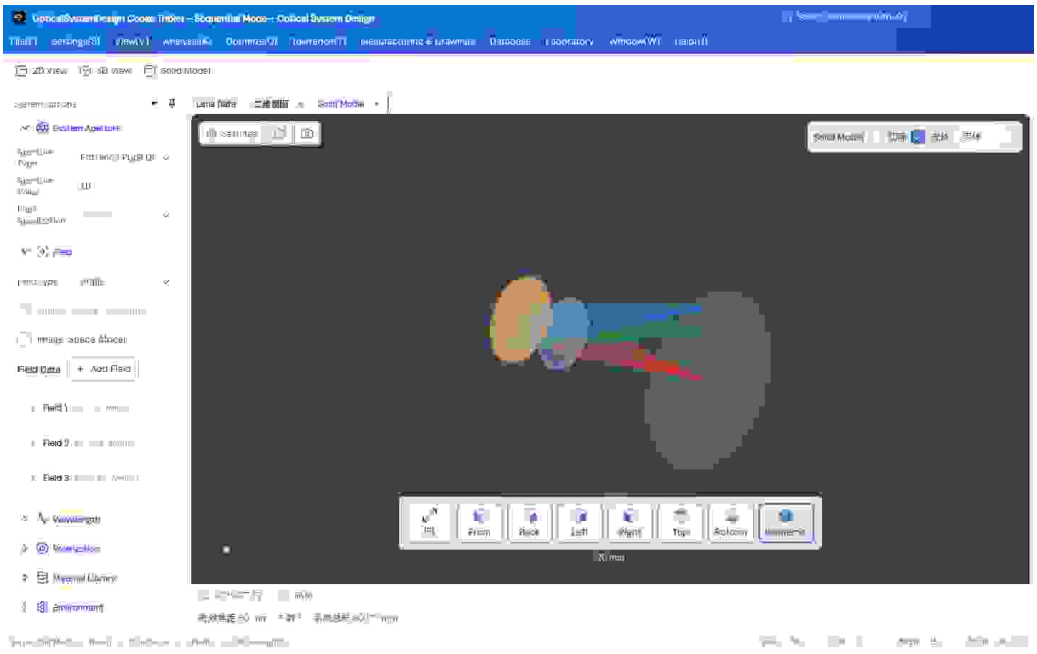
Use the ray equation below, solve the surface equation, then test propagation direction, domain and clear aperture. Spherical, aspheric and freeform surfaces use their actual geometry models; iterative intersections check convergence and validity. Missed, obscured or clipped rays are failures, not invented image points. General 3D surfaces are not replaced by a rotated 2D sketch.

$$\mathbf{p}(t) = \mathbf{o} + t\mathbf{d}$$
$$\mathrm{OPL} = \sum_j n_j(\lambda) L_j$$

Interfaces and total internal reflection

Apply vector refraction/reflection using the local normal. Snell's law sets the refracted angle; if no real refracted direction exists from high to low index, use total internal reflection. Folded systems preserve actual propagation direction rather than inferring it from Z order. OPL accumulates n times segment length; supported GRIN paths integrate the spatial medium rather than using one fixed index.

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$
$$\mathbf{d}_r = \mathbf{d} - 2(\mathbf{d}\cdot\mathbf{N})\mathbf{N}, \quad |\mathbf{d}| = |\mathbf{N}| = 1$$



Solid Model and view presets

How calculations work

First-order optics and focal length

Paraxial matrices describe small-angle behavior, separately from full aberration tracing.

Height–slope matrices

The centered homogeneous fixed-plane matrix uses height and geometric slope, not reduced angle. The equations give translation, refraction and the applicable air-to-air focal length. Successive stages left-multiply. Other endpoint media and reference planes require their actual definitions.

$$\begin{pmatrix} y_2 \\ u_2 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} y_1 \\ u_1 \end{pmatrix}, \quad u = \frac{dy}{dz}$$

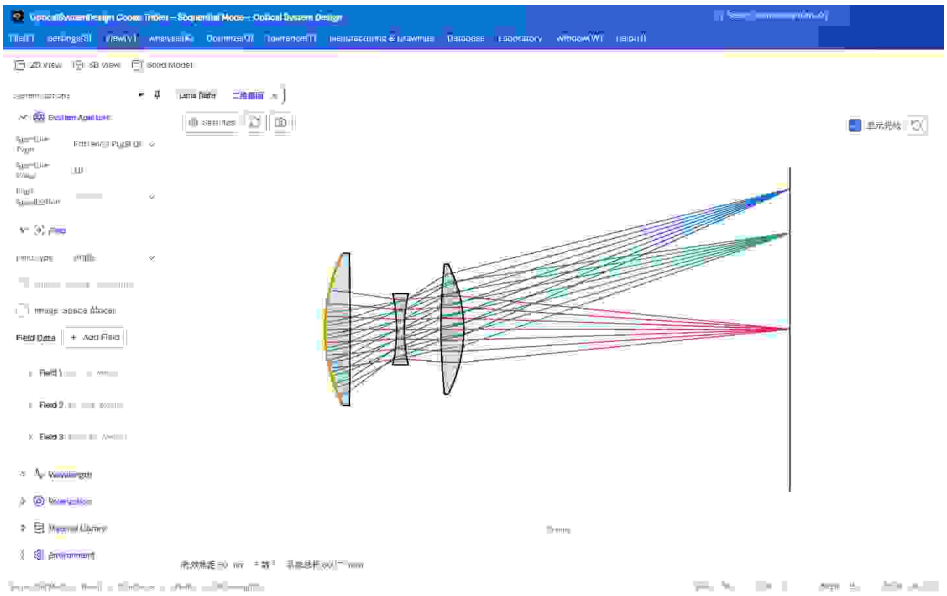
$$T(d) = \begin{pmatrix} 1 & d \\ 0 & 1 \end{pmatrix}$$

$$R = \begin{pmatrix} 1 & 0 \\ -\dfrac{n_2 - n_1}{n_2 R_s} & \dfrac{n_1}{n_2} \end{pmatrix}$$

$$\mathrm{EFL} = -\frac{1}{C} \quad \text{(air to air)}$$

Reading the summary

Focal length, back focus, principal planes and F/# summarize the system; they do not prove image quality across the field. Decenter, tilt, non-paraxial rays, reflection and complex media have entry-specific limits. Do not substitute a thin-lens estimate for an explicitly unsupported calculation. Continue with spot, wavefront and MTF analyses.



Inspecting the 2D layout

Y-Ybar and cardinal reports

Y-Ybar plots paraxial chief height horizontally and marginal height vertically at each surface. For image-height fields, compatible mode scales every chief height by the shown image-height factor. Cardinal reports use the first-order service; paraxial reference planes/focus are not real-ray best focus.

$$\bar{y}_j^{\text{scaled}} = \bar{y}_j \frac{h_{\text{selected}}}{\bar{y}_{\text{image}}}, \quad \bar{y}_{\text{image}} \neq 0$$

How calculations work

Spot, RMS and aberrations

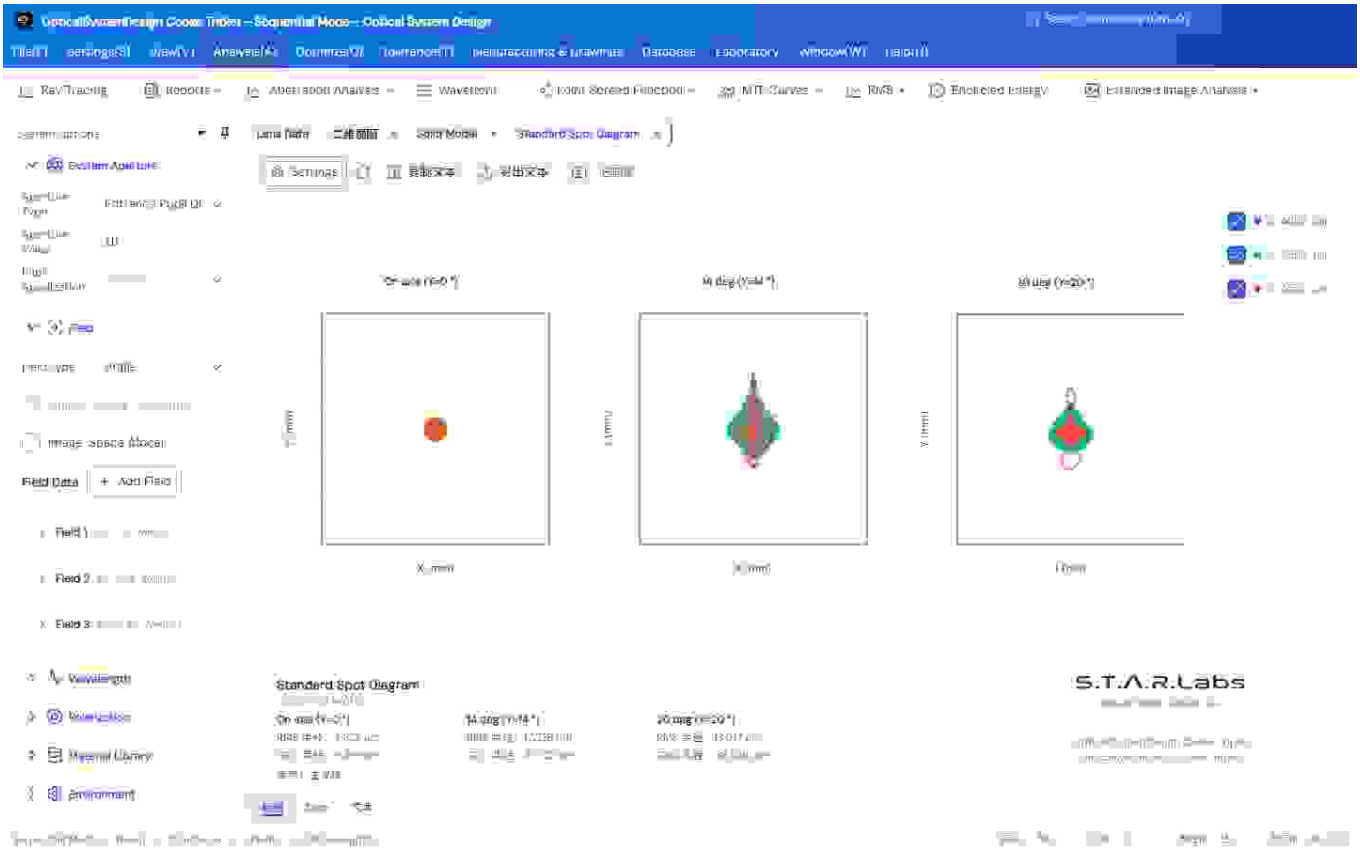
Turn traced image points into interpretable geometric image-quality metrics.

Centroid and RMS radius

A spot uses valid evaluation-surface intersections for the chosen field, wavelengths and pupil samples. A centroid reference computes the weighted centroid and weighted mean squared distance. Chief-ray and other references are different definitions. Failed samples, vignetting and weights follow the analysis contract; compare transmission, sampling and units before comparing RMS.

$$\bar{x} = \frac{\sum_i w_i x_i}{\sum_i w_i}, \quad \bar{y} = \frac{\sum_i w_i y_i}{\sum_i w_i}$$

$$\mathrm{RMS} = \sqrt{\frac{\sum_i w_i [(x_i - \bar{x})^2 + (y_i - \bar{y})^2]}{\sum_i w_i}}$$



Spot diagram: fields, wavelengths and results

Aberration is not energy

Ray fans show transverse error against pupil position; field curvature and distortion vary across the field. Seidel terms are a third-order paraxial expansion, not a replacement for full tracing at strong

aspheres or large fields. Geometric energy counts weighted rays; diffraction energy integrates image-plane PSF intensity. Hiding a legend series changes visibility, not the definition.

How calculations work

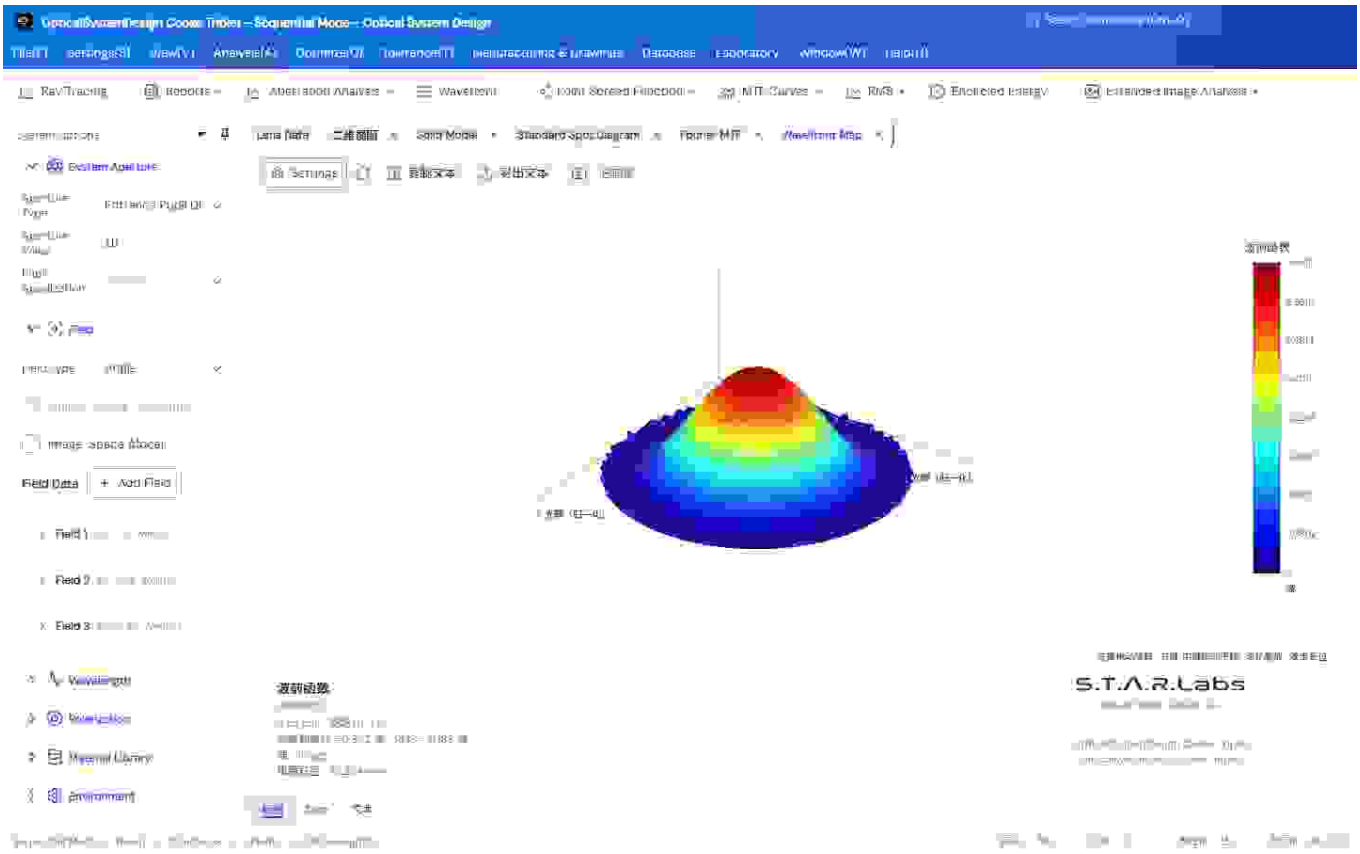
Wavefront, OPD and PSF

Build complex amplitudes from optical path differences and form image intensity.

The reference determines OPD

The project's OPD is reference minus actual path. The equation shows the FFT's negative pupil-phase convention paired with its transform convention; do not change one sign alone. Keep wavelength units consistent for waves, mm or µm. Chief reference removes piston; centroid reference can remove weighted best-fit tilt. Match pupil validity and sampling when comparing PV/RMS.

$$\varphi = -\frac{2\pi\,\mathrm{OPD}}{\lambda}$$
$$P(\xi,\eta) = A(\xi,\eta)\exp[i\varphi(\xi,\eta)]$$



Wavefront map and reference

FFT and Huygens

FFT PSF Fourier-propagates a complex pupil using the defined pupil and physical image sampling. Supported Huygens calculations sum sampled contributions with actual propagation distance, phase and weights. The equations show the core relation; mapping, weights and normalization remain part of the implementation. Different wavelengths normally add intensities incoherently on one physical image grid. Polarization, coherent beams and configurations require the entry's stated support.

$$I_{\mathrm{FFT}} \propto \left|\mathcal{F}\{P\}\right|^2$$

$$I_{\mathrm{Huygens}} \propto \left|\sum_j a_j e^{ik\,\mathrm{OPL}_j}\right|^2$$

$$I_{\mathrm{poly}}(x,y) = \frac{\sum_\lambda w_\lambda I_\lambda(x,y)}{\sum_\lambda w_\lambda}$$

How calculations work

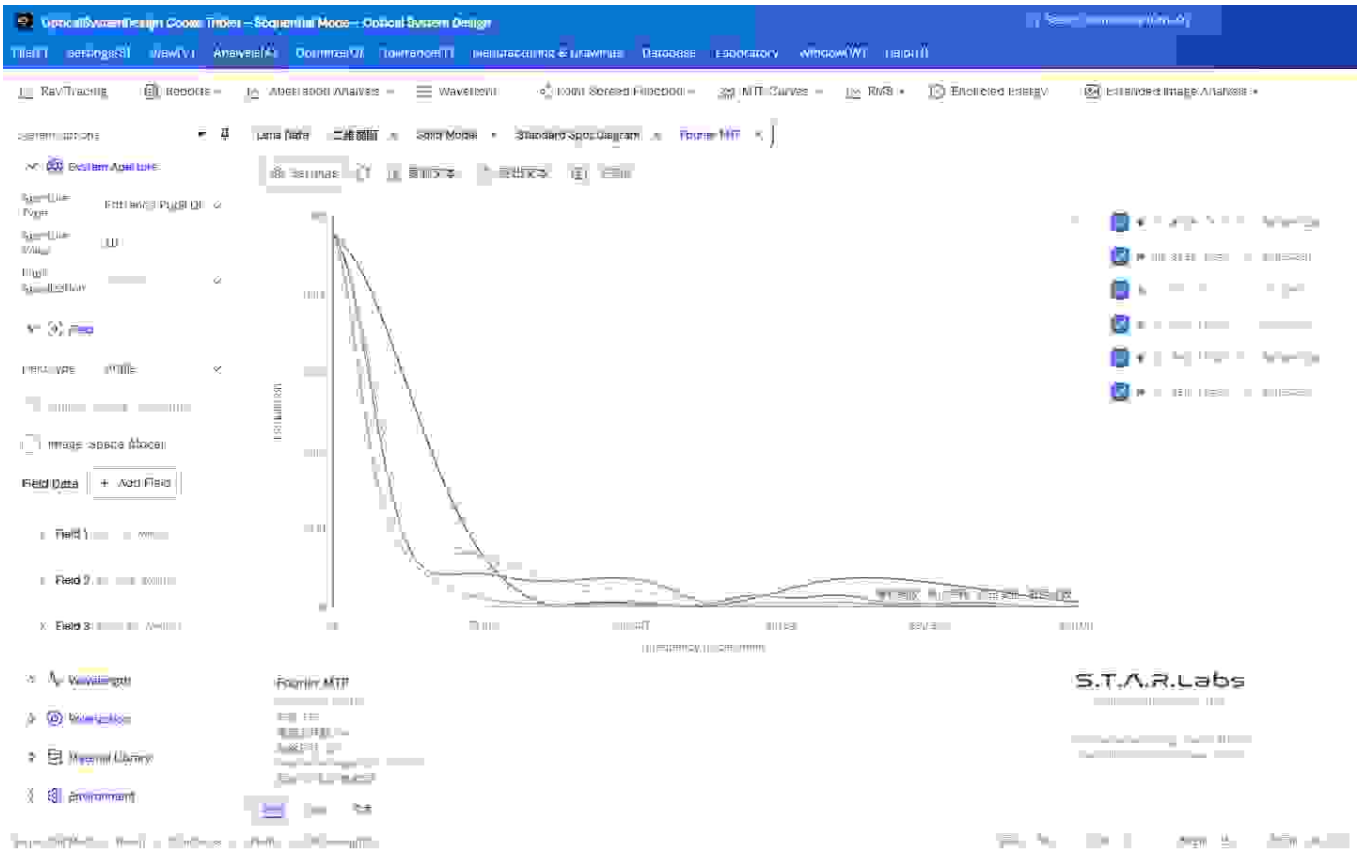
MTF, resolution and encircled energy

Convert PSFs or geometric image points to spatial-frequency and energy metrics.

OTF and MTF

Diffraction OTF comes from the Fourier transform of the PSF or equivalent pupil autocorrelation. MTF is its zero-frequency-normalized magnitude. Frequency is commonly cycles/mm and must match image-plane length units. Tangential/sagittal curves are directional sections. Geometric MTF uses image points without the complete diffraction limit; sampled MTF includes its sampling model. These entries are not interchangeable algorithms.

$$\mathrm{OTF}(f_x, f_y) = \mathcal{F}\{I(x, y)\}$$
$$\mathrm{MTF}(f_x, f_y) = \frac{|\mathrm{OTF}(f_x, f_y)|}{|\mathrm{OTF}(0, 0)|}$$



Fourier MTF curves

Energy integration and sampling

Diffraction encircled energy integrates the PSF about the selected center and normalizes by the defined total. Center, image window, pitch and integration range matter. Increase window and sampling to check convergence before interpreting high-percentile radii. An Airy disk assumes an ideal circular pupil and does not replace obscured, aberrated, rectangular or polychromatic calculations.

$$\mathrm{EE}(r) = \frac{\displaystyle\iint_{(x-x_c)^2+(y-y_c)^2 \le r^2} I(x,y)\,dx\,dy}{\displaystyle\iint I(x,y)\,dx\,dy}$$

Geometric versus sampled MTF

The expressions show geometric image-point OTF and current Sampled MTF's pupil phase-difference sum. Match position/frequency units. Sampled MTF fits a Fringe wavefront, maps frequency to pupil shear, integrates over the overlap and normalizes at zero, not an arbitrary pixel-sinc multiplier. W is in waves and Δ is normalized pupil shear.

$$\mathrm{OTF}_{\mathrm{geometric}}(\mathbf{f}) = \frac{\sum_j w_j \exp[-2\pi i\,\mathbf{f}\cdot\mathbf{x}_j]}{\sum_j w_j}$$

$$Q(\Delta) = \sum_{p \in \mathcal{P}\cap(\mathcal{P}+\Delta)} I_p e^{2\pi i[W(p)-W(p-\Delta)]}$$

$$\mathrm{OTF}_{\mathrm{sampled}}(\Delta) = \frac{Q(\Delta)}{Q(0)}$$

Contrast loss map

For a selected field, wavelength and frequency, Moore–Elliott sheared pupil-ray pairs give common-reference wavefront differences and local pupil loss. Invalid pairs stay unavailable. Optional display normalization divides by the map's maximum loss. This is not a field-wide imaging MTF.

$$L(p) = \frac{1-\cos(2\pi\Delta W(p))}{2}$$

How calculations work

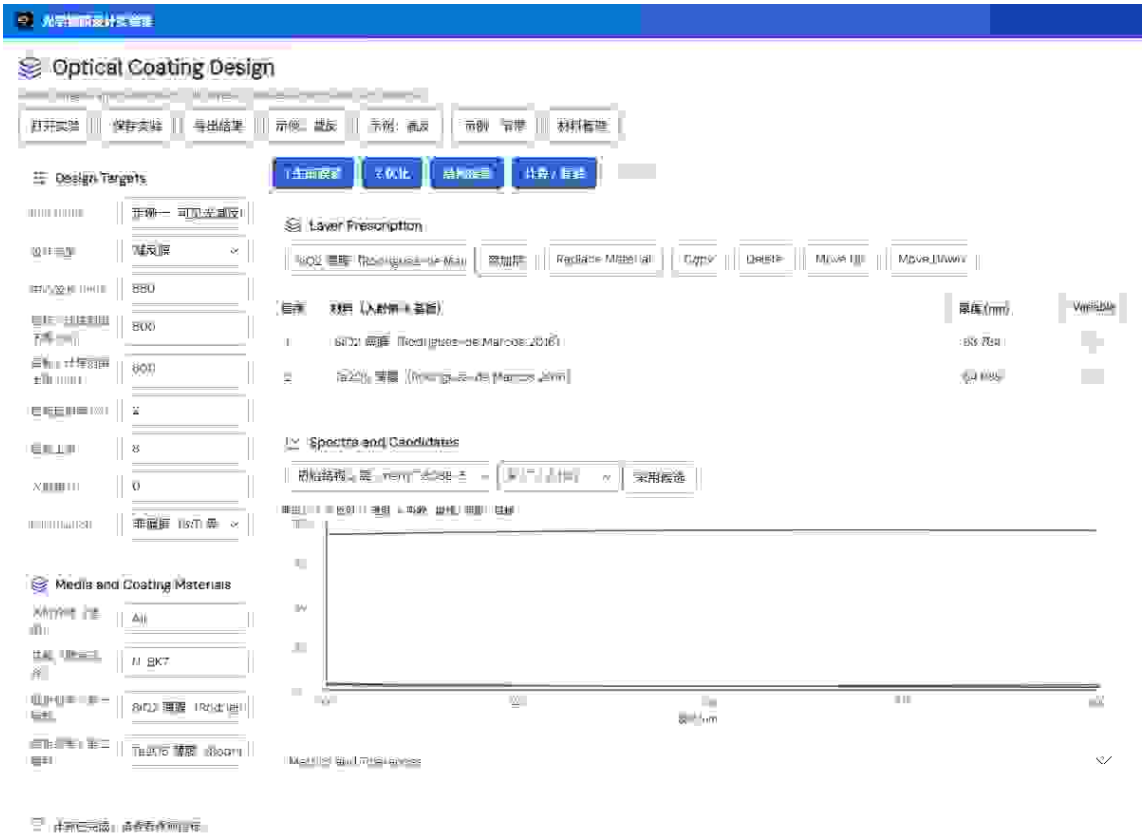
Polarization, Fresnel and coatings

Distinguish field amplitudes, intensity and power at optical interfaces.

Interfaces and coherent layers

Bare-interface S/P Fresnel coefficients depend on angle, wavelength and both media. Films use complex indices and physical thicknesses to calculate interference and R/T/A; absorption must be included when interpreting transmittance. Polarized tracing retains Jones fields and reference bases. Unpolarized models normally average powers from orthogonal inputs rather than averaging electric fields.

$$R = |r|^2$$
$$R + T + A = 1$$



Coating design and spectrum

Jones-pupil reference bases

Jones Pupil propagates two independent unit inputs along each real path to form a complex matrix between orthonormal transverse bases. Entries are amplitudes/phases, not four power transmittances. This entry has separate Fresnel/bulk-absorption settings and rejects physical multilayer coatings instead of substituting bare interfaces.

$$\begin{pmatrix} E_x^{\mathrm{out}} \\ E_y^{\mathrm{out}} \end{pmatrix} = \begin{pmatrix} J_{xx} & J_{xy} \\ J_{yx} & J_{yy} \end{pmatrix} \begin{pmatrix} E_x^{\mathrm{in}} \\ E_y^{\mathrm{in}} \end{pmatrix}$$

Scope and boundaries

Wavelength ranges, absorption, exterior media, GRIN and coherent XYZ propagation have separate limits. Transparent homogeneous support does not imply arbitrary absorbing-medium vector fields. Respect explicit unsupported states instead of substituting an empirical transmission factor.

How calculations work

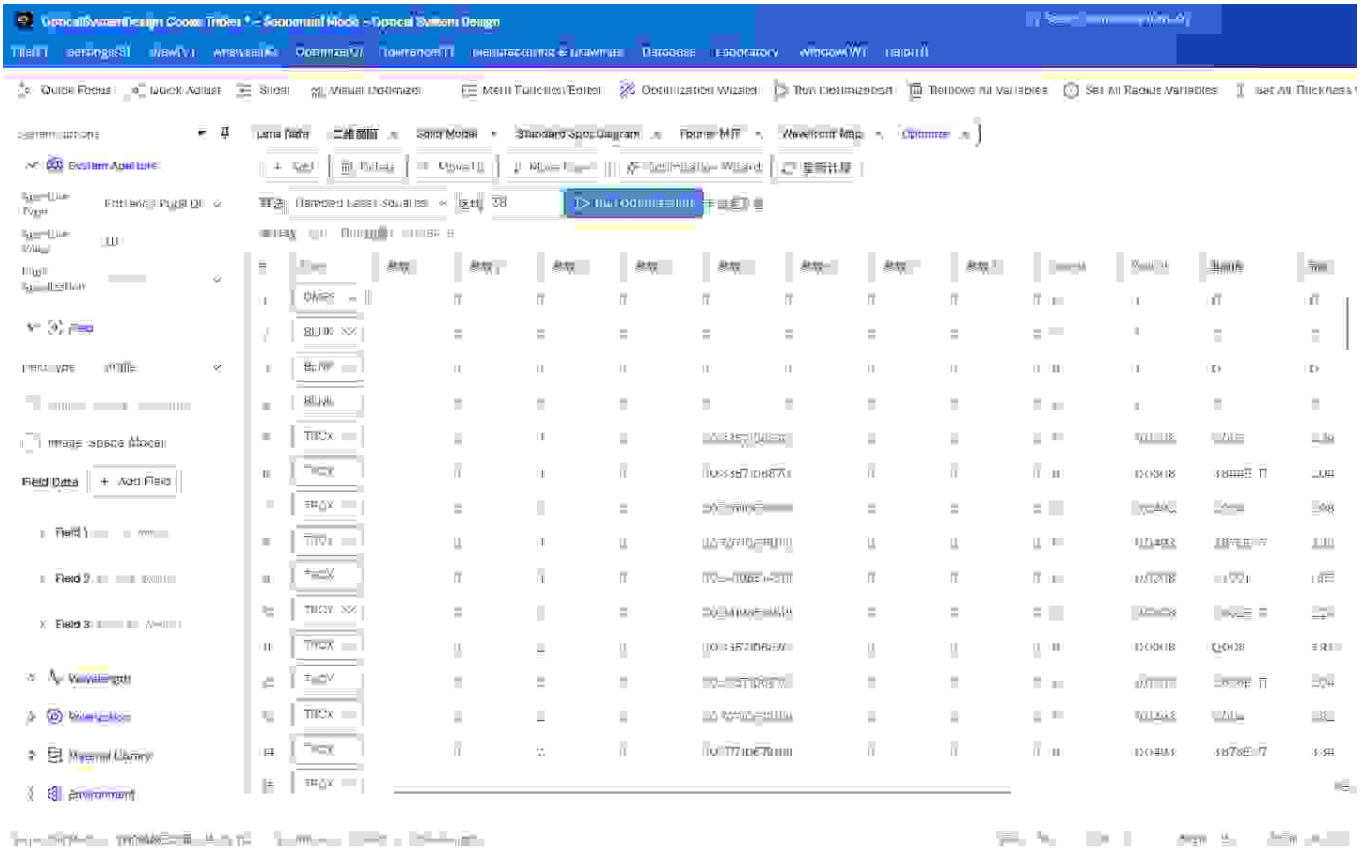
Merit functions and optimizers

Understand variables, residuals and stopping conditions without treating success as certification.

Variables, targets and weights

Variables have bounds and step/scaling settings. General curvature variables use 1/mm internally, with zero for a plane. Positive weights create objective residuals, negative weights create constraint residuals and zero weights monitor only. The common problem normalizes by the sum of absolute weights; objective merit and constraint error are separate. Display contribution normalization is not another optical metric.

$$r_i = \sqrt{\frac{w_i}{\sum_j |w_j|}}(v_i - t_i), \quad w_i > 0$$
$$\text{merit} = \sum_{i:w_i>0} r_i^2$$



Merit Function Editor

Algorithms and stopping reasons

DLS uses a damped residual Jacobian and its supported constraint handling. Nelder–Mead reflects, expands, contracts inside/outside and shrinks a simplex. Coordinate search and seeded random perturbation use different trials. Reaching a limit or improving merit does not prove a global optimum. Inspect constraints, transmission, manufacturability and independent analyses. Results commit only if the source project has not changed concurrently.

How calculations work

Sensitivity, Monte Carlo and yield

Follow manufacturing perturbations into image-quality metrics, compensation and yield.

Sensitivity and compensation

Sensitivity applies each lower/upper endpoint to the nominal design, retraces the selected criterion, optionally optimizes compensators and restores nominal state. RSS is a local statistical estimate, not a replacement for simultaneous-error Monte Carlo. Compensators are allowed adjustments, not sampled manufacturing errors.

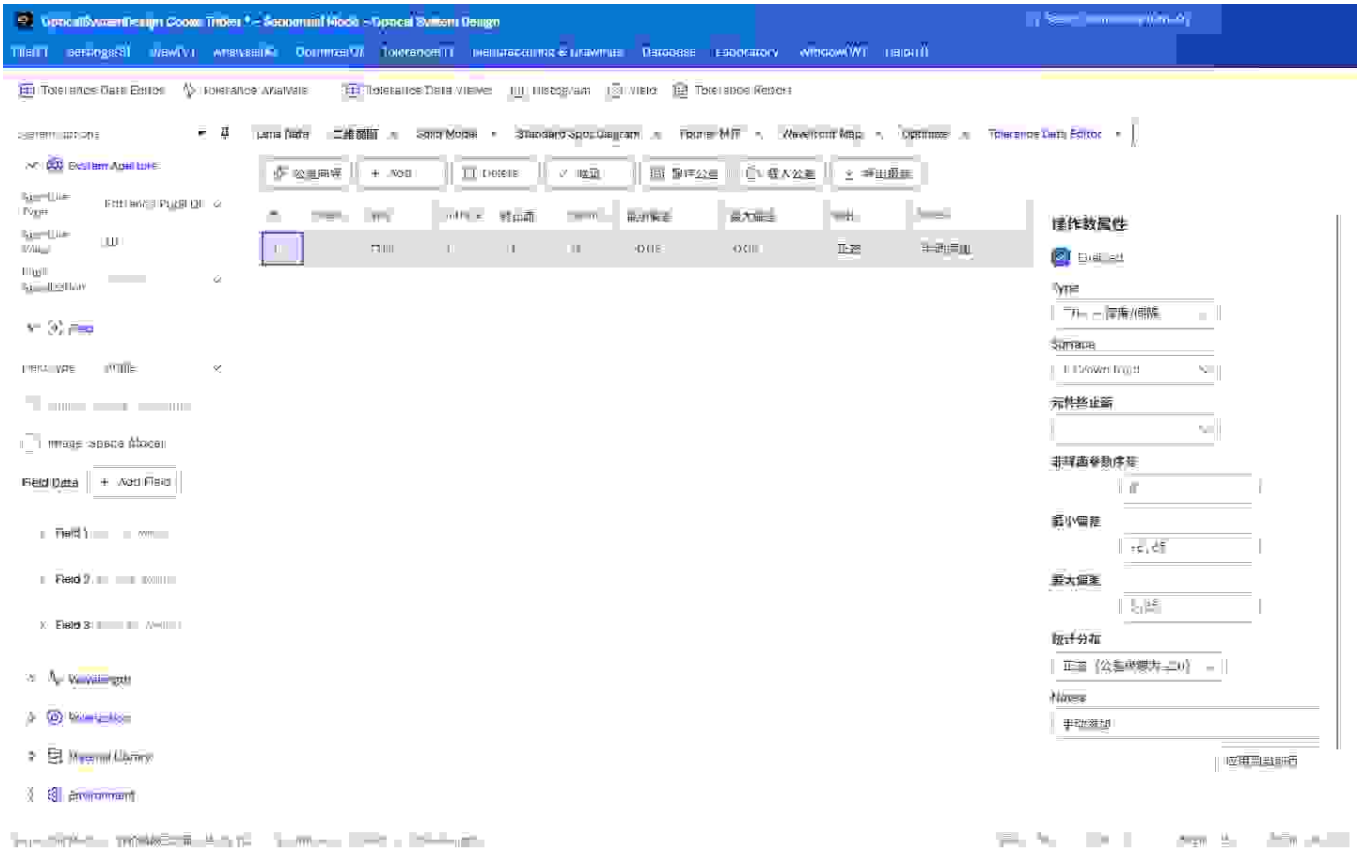


Tolerance analysis settings

Deterministic trials and yield

Each trial starts from a nominal snapshot, simultaneously samples the defined distributions and evaluates uncompensated/compensated criteria. Seeds derive from trial index so serial and parallel results keep the same order. Yield checks all configured acceptance conditions; failed trials cannot be silently removed to improve yield. It estimates the chosen model/distributions, not measured production yield.

$$\widehat{Y} = \frac{N_{\text{pass}}}{N_{\text{trials}}}$$



Tolerance Data Editor

How calculations work

Non-sequential paths, power and detectors

Find the next spatial interaction rather than the next surface number.

Objects and branches

Sources define position, direction, wavelength and power. A BVH narrows object candidates and selects the nearest forward hit. Geometry determines the initial medium inside solids. Splitting creates transmitted/reflected branches; segment, active-branch and relative-intensity limits truncate paths. Truncated power is reported separately, not counted as detected or absorbed.

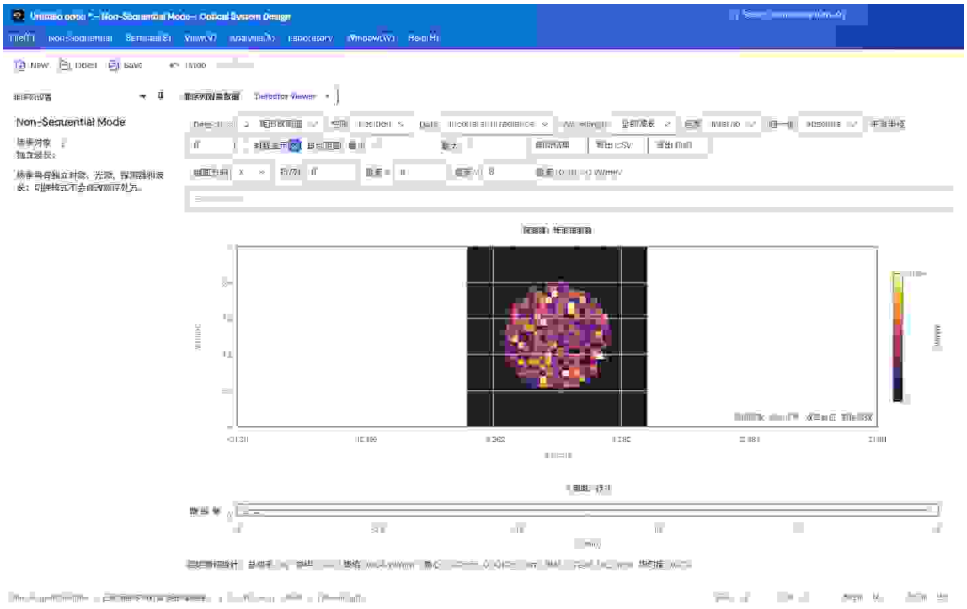
$$P_{\text{source}} = P_{\text{detector}} + P_{\text{absorbed}} + P_{\text{escaped}} + P_{\text{truncated}}$$

Pixel quantities and databases

Pixel power sums accepted ray powers. Position-space irradiance divides by physical pixel area with the stated unit conversion; angle-space intensity follows its angular-bin definition. Unknown wavelengths remain unknown and physical edges map to real pixels. Database reconstruction checks scene/material identity; retrace after geometry/material changes. Clear and Trace resets totals; Trace Only accumulates compatible sessions.

$$P_{\text{pixel}} = \sum_i P_i$$

$$E_{\text{pixel}} = \frac{P_{\text{pixel}}}{A_{\text{pixel}}}$$



Detector results after tracing

How calculations work

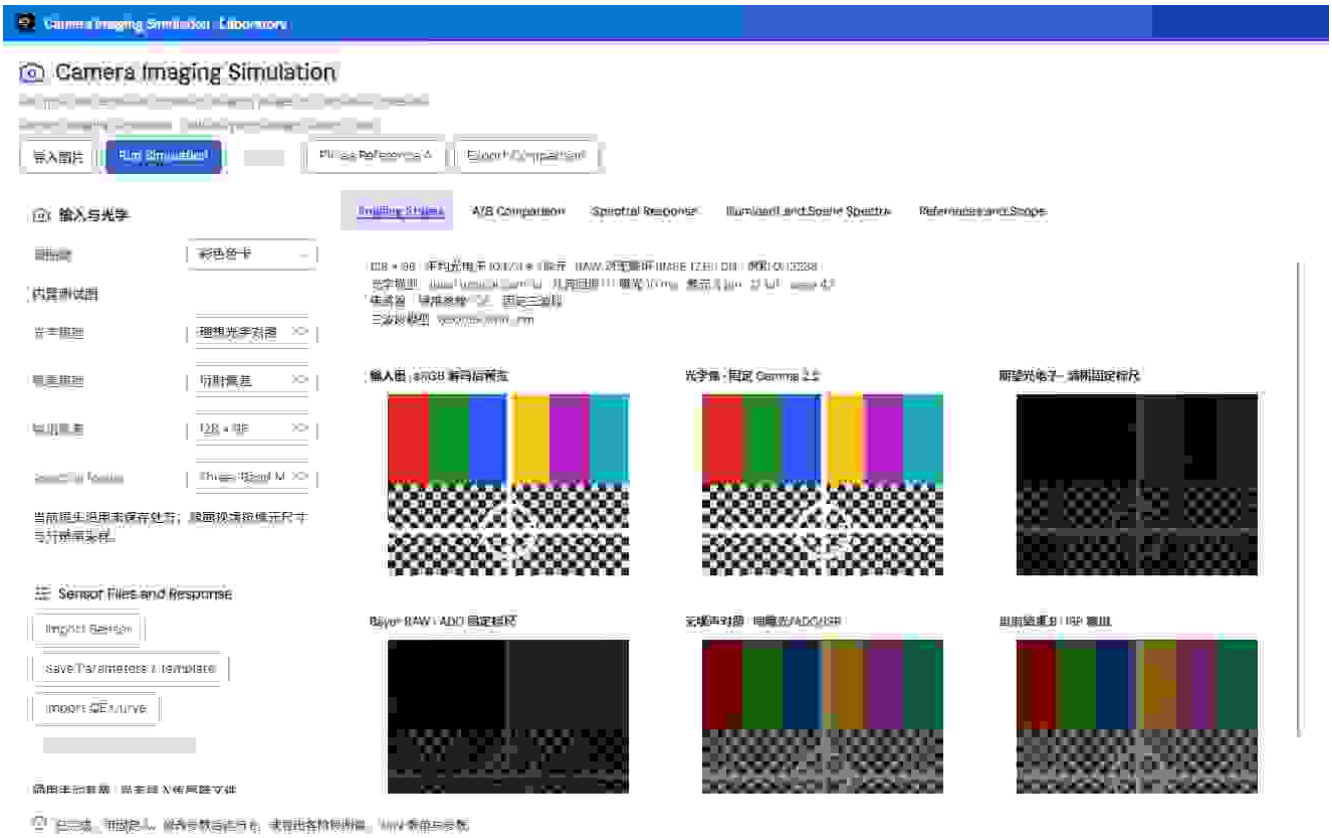
How laboratories share calculations

The laboratories have different jobs while sharing the physical optical core.

Camera: from irradiance to electrons

Current Lens uses a snapshot for spatial PSF, distortion and illumination; Ideal Optics is a separate control. L is linear image value, A is in m^2, t in s, wavelength in m, and Eref is full-white per-band irradiance in W/m^2. Absolute QE gives expected electrons before noise, dark current, nonuniformity, saturation, gain, black level, ADC and ISP. RGB is not a measured spectrum; continuous mode integrates supplied spectral curves.

$$N_e = E_{\mathrm{ref}}\, L\, A\, t\, \frac{\lambda}{hc}\, \mathrm{QE}$$



Camera imaging laboratory

Initial Structure, Coating and Assembly

Standard Initial Structure uses managed AutoLens to organize candidates, loss and optimization while sharing tracing, materials and formal acceptance. CUDA changes eligible backends and fallbacks remain disclosed. Coating organizes layer stacks and targets around the common film solver; Assembly uses its measurement/workflow model. Each result has its own state and must still be checked in the

main project.

How calculations work

Checking numerical reliability

Inspect settings, convergence, revisions and validation scope rather than appearance alone.

Repeatable comparisons

Hold prescription, configuration, materials, field, wavelengths, reference, polarization, sampling and normalization fixed. Increase pupil/image sampling or PSF window and check stability. Include failures, truncations, geometric fallbacks and unsupported states in interpretation rather than hiding failed samples.

Scope of verification

Model checks, independent equations, regressions, external captures and UI screenshots verify different things. The project uses captured Zemax OpticStudio 2026 R1 cases as its primary external numerical reference, limited to those files/settings. This does not establish universal equivalence. Screenshots explain operation, not numerical accuracy.

How calculations work

Surfaces, coordinate transforms and GRIN

Intersections use actual surface equations; graded index requires path integration.

Conics and even aspheres

The sag expression describes a rotational conic with even asphere terms. Lengths share geometry units; curvature is reciprocal radius only for a finite nonzero radius, and zero for a plane. Check the radical domain, aperture and pose. XY, biconic, toroidal, Zernike and grid surfaces use their own models. Transform local points/normals with the complete rigid pose.

$$z(r) = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2r^2}} + \sum_{m=2}^{M} A_{2m} r^{2m}, \quad c = \frac{1}{R}$$

Spatially varying media

GRIN bends rays continuously with the index gradient. The arc-length equation describes geometric propagation; implementation also handles integration steps, boundary events and failures. Check step convergence separately from pupil quadrature. Unsupported media/coordinate combinations must be rejected rather than replaced with a constant index.

$$\frac{d}{ds}\left(n\frac{d\mathbf{r}}{ds}\right) = \nabla n$$

How calculations work

Pupil sampling, aiming and vignetting

Positions, weights and validity define quadrature; ray count alone is not accuracy.

Quadrature weights

Rectangular, hexapolar and other samplings have different nodes and area weights. The equation illustrates a weighted mean; each analysis chooses launch/transmission weights and normalization. Renormalizing surviving rays can hide vignetting. Boundaries, obscuration, spectral weights and duplicate nodes affect integration.

$$\langle g \rangle \approx \frac{\sum_i w_i g_i}{\sum_i w_i}, \quad \sum_i w_i \longrightarrow \int_{\mathcal{P}} dA$$

Aiming and failed samples

Aiming solves for the actual requested stop coordinate; entrance-pupil coordinates are not universal stop intersections. Chief, boundary and integration rays serve different roles. Preserve miss, TIR, clipping and aiming-failure reasons. CPU/GPU comparisons require identical nodes, precision, aiming policy and validity checks.

How calculations work

Seidel, field curvature, distortion and color

Distinguish third-order coefficients, real-ray focus and image-height errors.

Seidel coefficient conventions

The expression illustrates the third-order wave-aberration basis, not a sum of the report's seven columns. Reports distinguish Seidel values in mm, wavelength-scaled wave coefficients, and transverse/longitudinal aberrations. Conversion depends on image index, paraxial marginal slope and wavelength. Sum surface contributions only with consistent bases and units.

$$W(\rho,\theta;H) = W_{040}\rho^4 + W_{131}H\rho^3\cos\theta + W_{222}H^2\rho^2\cos^2\theta + W_{220}H^2\rho^2 + W_{311}H^3\rho\cos\theta$$

Real rays and reference height

Field curvature compares tangential/sagittal focus with the reference image plane. Distortion compares actual chief-ray height with the selected reference, using a defined on-axis treatment instead of division by zero. Color focus scans wavelength, lateral color compares image heights, and axial aberration scans pupil intersections. Grid distortion extends the mapping to 2D.

$$D = 100\,\frac{h_{\text{real}} - h_{\text{reference}}}{h_{\text{reference}}}\,\%, \quad h_{\text{reference}} \neq 0$$

How calculations work

How Zernike coefficients are fitted

Basis and sampling domain define coefficients; numbering is not interchangeable.

Design matrix and QR least squares

Build the design matrix from Zernike basis values at valid pupil nodes and OPD in waves. Current coefficient fitting uses unweighted QR least squares on those nodes. Intensity-weighted wavefront statistics are a separate definition. Check conditioning, rank, term/resource limits and fit residuals.

$$W(\rho,\theta) \approx \sum_{j=1}^{K} a_j Z_j(\rho,\theta)$$

$$\mathbf{a} = \underset{\mathbf{a}}{\operatorname{argmin}} \ \|Z\mathbf{a} - \mathbf{W}\|_2^2$$

Fringe, Standard and Annular

Fringe and Standard use different numbering/normalization. Annular also depends on obscuration and an annular orthogonal basis. Compare only with matching obscuration, domain, reference sphere, piston/tilt removal and sampling. Coefficients-vs-field refits at each sampled field. The retired generic Zernike page is not restored.

How calculations work

Interferogram phase and intensity

Difference two wavefronts, then apply scale, chief path and pupil tilt.

The fringe equation

Wavefronts are in waves at the selected wavelength; the reference beam has zero wavefront error. Scale is s; tx/ty count fringes across a full pupil diameter and are added after scaling. Include the chief-path term only when enabled, with consistent length units. The project's OPD is reference minus actual path, giving the shown chief-path sign.

$$q = s\left(W_1 - W_2 + \frac{\mathrm{OPL}_2 - \mathrm{OPL}_1}{\lambda}\right) + \frac{t_x x + t_y y}{2}$$

$$I = \frac{1 + \cos(2\pi q)}{2}$$

Valid pupil and display

The model assumes equal-strength beams with fully overlapping normalized pupils and uses common valid traced samples. A grayscale palette can map intensity one to black without reversing physical data. Subapertures, small-angle vertex reference and intermediate surfaces have explicit limits. An OPD heatmap is not this interferogram.

How calculations work

Foucault knife edge and angular spectrum

Propagate a masked complex field and distinguish displayed intensity from total throughput.

Focal knife and ideal pupil reimaging

The default FFTs the shared complex pupil, applies a focal knife mask, then inverse-transforms to an ideal reimaged pupil. Supported polarization channels propagate separately before power summation; unpolarized input states are incoherent. Padded-grid power, central displayed power and knife throughput are distinct. Do not remask diffracted spill outside the original circle.

$$U_{\text{out}} = \mathcal{F}^{-1}\{M\,\mathcal{F}[P]\}$$
$$I = \sum_c |U_{\text{out},c}|^2, \quad \tau = \frac{P_{\text{out}}}{P_{\text{in}}}$$

Spatial and vector angular spectrum

Angular-spectrum modes use physical knife/observation distances and window extent. The scalar relation is shown; vector mode also applies transverse projection and medium restrictions, not arbitrary-media vector POP. Check sampling and extent together. A focal knife in micrometers requires defined focusing optics.

$$U(z) = \mathcal{F}^{-1}\left[\mathcal{F}[U(0)]\exp\left(iz\sqrt{k^2 - k_x^2 - k_y^2}\right)\right]$$

How calculations work

Relative illumination and radiometry

Identify relative normalization and the area/solid-angle units of absolute power.

Relative illumination is not a fixed cosine-fourth curve

The shared integral evaluates image-side projected cosine pupil area Q using actual transmission geometry and selected sampling/vignetting/polarization settings. The current curve divides by the maximum Q of the sampled scan, not necessarily its on-axis value. Record mode, actual nodes and whether scalar transmission is included.

$$\mathrm{RI}(H) = \frac{Q(H)}{\max\limits_{H' \in \mathcal{H}_{\mathrm{sampled}}} Q(H')}$$

Irradiance and radiant intensity

Incoherent irradiance bins image-plane power per pixel area; radiant intensity bins direction per solid angle. Absolute units are W/mm² and W/sr respectively; normalized output is not absolute power. Geometric statistics do not add arbitrary coherent interference. Preserve source power, quadrature, vignetting and transmission conventions.

$$E_{ij} = \frac{\Phi_{ij}}{\Delta A_{ij}} \quad [\mathrm{W/mm^2}]$$

$$J_{ij} = \frac{\Phi_{ij}}{\Delta \Omega_{ij}} \quad [\mathrm{W/sr}]$$

How calculations work

Line spread, edge spread and extended sources

A cross section, projected integral and cumulative integral are different operations.

LSF and ESF

Diffraction LSF integrates the 2D PSF in the perpendicular direction; ESF cumulatively integrates and normalizes it. A central PSF slice is generally not an LSF. Geometric line/edge spread instead projects and accumulates weighted ray distributions. Record direction, center, units, window and normalization.

$$\mathrm{LSF}(x) = \int \mathrm{PSF}(x, y)\, dy$$

$$\mathrm{ESF}(x) = \frac{\int_{-\infty}^{x} \mathrm{LSF}(u)\, du}{\int_{-\infty}^{\infty} \mathrm{LSF}(u)\, du}$$

Extended-source energy

Extended-source energy combines the source distribution with its chosen geometric/diffraction response before integration. Source size/shape, image scale and reference affect the result. A point-source Airy radius is not directly comparable to an extended-source percentile. Check truncation and image extent.

How calculations work

Geometric, coherent and partially coherent images

Distinguish intensity and field addition before checking spatial variation and sampling.

From point responses to images

Geometric imaging maps source samples through real rays and bins power. Incoherent diffraction adds intensity responses; coherent imaging adds complex amplitudes before squaring. The equations retain source-dependent response; convolution assumes shift invariance. Bitmap and IMA/BIM viewers display files without new optical calculations.

$$I_{\text{incoherent}}(\mathbf{x}) = \int I_o(\mathbf{u}) h_I(\mathbf{x};\mathbf{u})\, d\mathbf{u}$$
$$U_{\text{coherent}}(\mathbf{x}) = \int U_o(\mathbf{u}) h_U(\mathbf{x};\mathbf{u})\, d\mathbf{u}$$
$$I_{\text{coherent}}(\mathbf{x}) = |U_{\text{coherent}}(\mathbf{x})|^2$$

Mutual intensity and scope

Partial coherence propagates source mutual intensity with a two-point kernel, not an arbitrary blend of sharp/blurred images. The implemented source/coherence model has defined construction, positivity and resource bounds. Phase, polarization and spatial propagation combinations remain scoped. Image results also depend on scale, orientation, boundaries and normalization.

$$I(\mathbf{x}) = \iint J_o(\mathbf{u}_1,\mathbf{u}_2) h_U(\mathbf{x};\mathbf{u}_1) h_U^*(\mathbf{x};\mathbf{u}_2)\, d\mathbf{u}_1\, d\mathbf{u}_2$$
$$J_{ij} = (1-\gamma) I_i \delta_{ij} + \gamma a_i a_j^*, \quad 0 \le \gamma \le 1$$

How calculations work

Scans, reference spheres and configurations

Scan nodes and reference definitions are calculation settings too.

Field, wavelength and focus scans

RMS spot vs field uses Field Density plus one positions from zero to the maximum defined field in the chosen direction, not just Field Editor rows. Other scans use their own nodes. Focus scans change evaluation position in a copy. Angle/height scans through pupil and field have different horizontal variables. Matrices organize results without changing units.

$$H_i = \frac{i}{N} H_{\max}, \quad i = 0, 1, \ldots, N$$

Reference spheres and configuration identity

Chief-ray, centroid and best-fit spheres use different centers/radii for OPD. Sphere fitting is not merely plane removal. Configuration overlays retain each prescription/result identity. Coherent beams, polarization and Huygens combinations have specific support; ordinary overlays do not sum electric fields.

How calculations work

Optimizer steps, derivatives and bounds

Residuals, derivatives and the applied variables must agree.

Damped least squares

DLS solves a damped linearized residual problem; its scale matrix follows the implemented variable/diagonal strategy, with constraints and step acceptance. Nelder–Mead instead uses simplex reflection, expansion, inside/outside contraction and shrinkage. Evaluate the actual bounded coordinates, not a vertex inconsistent with the prescription.

$$(J^{\mathsf{T}}J + \mu D)\,\Delta\mathbf{x} = -J^{\mathsf{T}}\mathbf{r}$$

Differences and differentiable tracing

Central differences require valid samples on both sides. Bounds, clipping and topology changes need supported one-sided/failure handling. Differentiable tracing propagates derivatives through shared geometry/refraction within valid smooth branches; clipping/TIR switches are not arbitrary smooth operations. Validate gradients against same-definition differences and preserve CPU/GPU physics.

$$\frac{\partial r_i}{\partial x_j} \approx \frac{r_i(\mathbf{x} + h_j\mathbf{e}_j) - r_i(\mathbf{x} - h_j\mathbf{e}_j)}{2h_j}$$

How calculations work

Thin-film phase, reflection and transmission

Complex indices and stable recursion handle absorption, not an empirical curve.

Phase thickness and stable recursion

The shared solver recursively combines S/P admittances and complex phase from substrate to incident side. Attenuation and logarithmic transmission amplitudes stabilize absorption. The expressions show layer phase and reflection recursion with consistent lengths. Coherent amplitudes combine before boundary flux normalization.

$$\delta_j = \frac{2\pi}{\lambda}\tilde{n}_j d_j \cos\theta_j$$

$$r_j = \frac{r_{j,j+1} + r_{j+1}e^{2i\delta_j}}{1 + r_{j,j+1}r_{j+1}e^{2i\delta_j}}$$

Materials and experiment criteria

Index sign convention, forward root, incident medium and substrate affect flux direction. Experiment criteria span wavelength, angle, polarization and tolerance; inspect R, T, A and failed criteria separately. Simplified coating models must not be presented as the full coherent multilayer solver or measured stacks.

How calculations work

Initial-structure search and independent acceptance

Training loss, actual compute device and formal acceptance are different results.

Standard AutoLens updates

The standard entry uses managed AutoLens with its recorded curriculum, fixed samples and Adam updates. Formal tracing/derivatives remain shared Core. First/second moments are bias-corrected, with step and epsilon defined by the recorded algorithm. Module-specific RGB loss, green centroid and detached gradients are not generic spot RMS.

$$
\begin{aligned}
m_t &= \beta_1 m_{t-1} + (1-\beta_1) g_t \\
v_t &= \beta_2 v_{t-1} + (1-\beta_2) g_t^2 \\
\widehat{m}_t &= \frac{m_t}{1-\beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1-\beta_2^t} \\
x_{t+1} &= x_t - \alpha_t \frac{\widehat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}
\end{aligned}
$$

Candidates, topology and pass criteria

Independently validate transmission, focal length, aperture, image quality and manufacturability with original gates. Autonomous topology and standard AutoLens are different experiment protocols; retain reduction, compensation and budget records. A passing case does not establish universal success. Distinguish fallback, budget exhaustion, incomplete runs and actual GPU execution.

How calculations work

Assembly conjugates and reticle images

Conjugates use the paraxial limit of shared real rays; previews trace finite-aperture return images.

Paraxial conjugate position

Assembly builds shared real-ray residuals for a target vertex/curvature center, differentiates small height/slope probes and extrapolates to check convergence. Positions reference the first physical vertex. Degenerate derivatives can indicate infinity. Unsupported paths, no reflection and numerical failure are distinct, not zero positions.

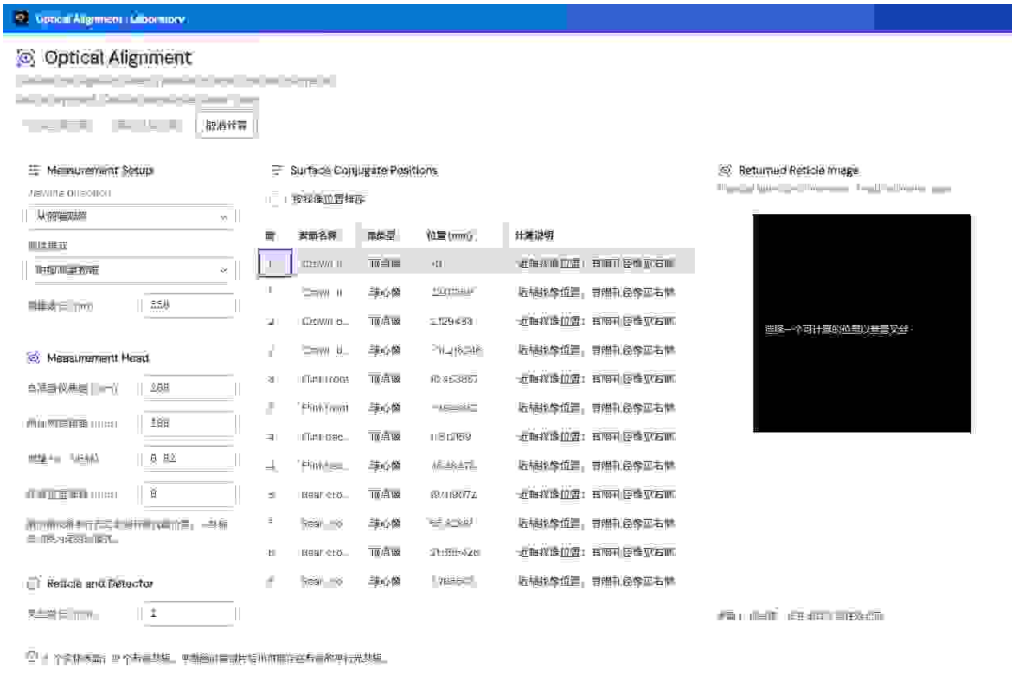
$$A = \frac{\partial f}{\partial h}, \quad B = \frac{\partial f}{\partial u}$$

$$D_{\text{ext}} = \frac{4D(h/2) - D(h)}{3}$$

$$z_{\text{image}} = z_{\text{launch}} + \sigma \frac{B}{A} - z_{\text{datum}}$$

Finite-aperture preview

Reticle previews use probe/collimator parameters and the selected reflecting surface, bin returned rays and report launched/returned/recorded counts. Focused and infinity modes differ. Brightness is normalized without a formal coating/stray-light power model. This is not a general sensitivity-matrix alignment inverse or an instrument measurement.



Optical Assembly workspace